

Module 1

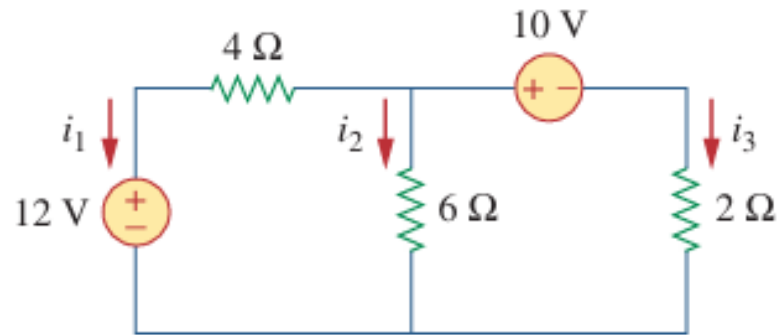
DC Circuit Analysis and Theorems

Dr. Elizabeth Rufus

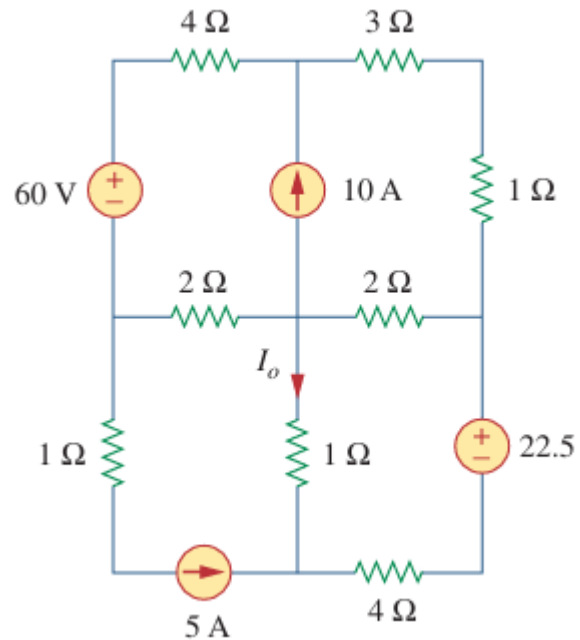
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Tutorial

1. Use mesh analysis to obtain i_1 , i_2 , and i_3 in the circuit

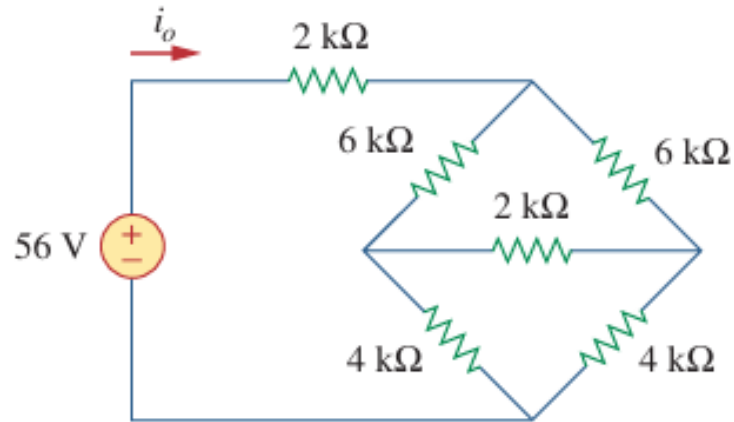


2. Apply mesh analysis to the circuit and obtain I_o .

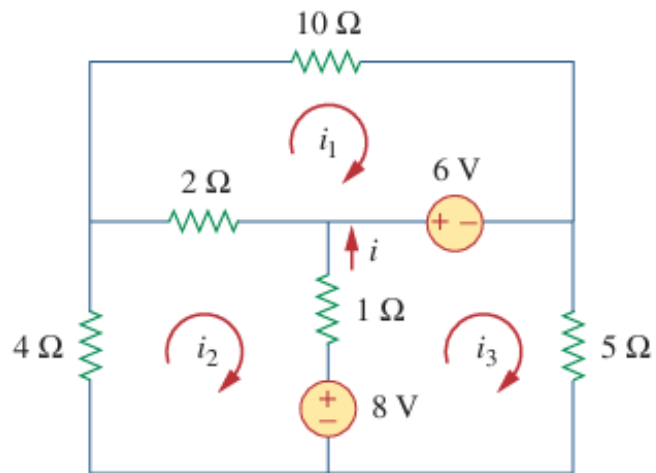


Tutorial

3. For the bridge network, find i_o using mesh analysis.

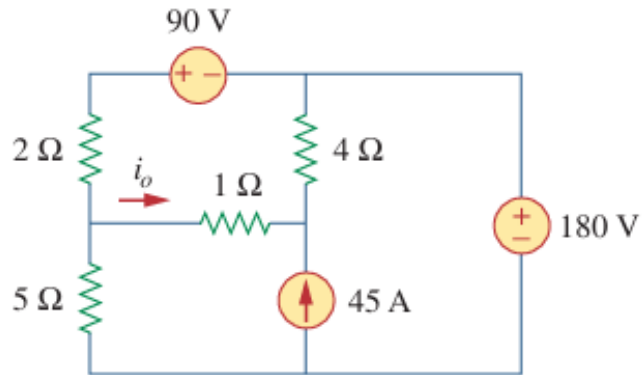


4. Apply mesh analysis to find i .

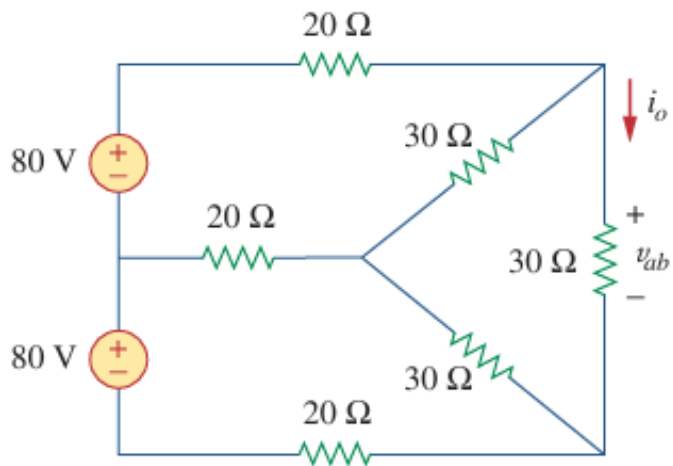


Tutorial

5. Use mesh analysis to obtain in the circuit.

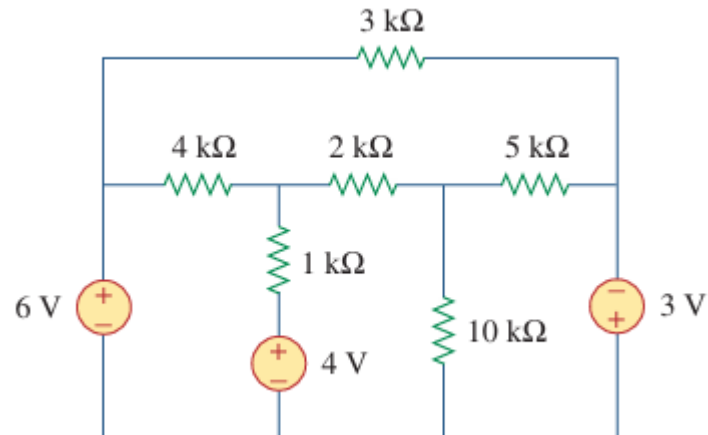


6. Use mesh analysis to find v_{ab} .

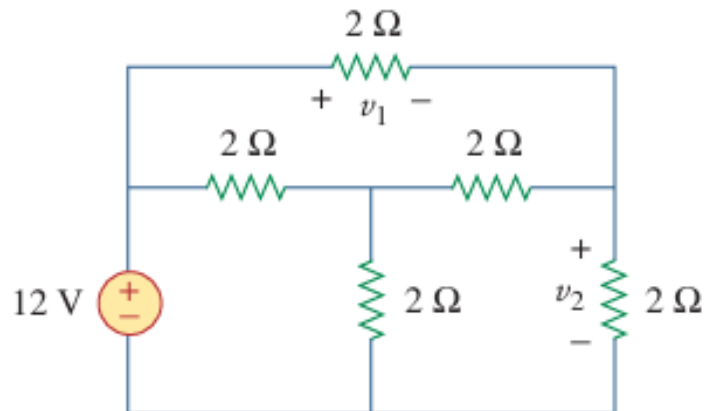


Tutorial

7. Determine the current through the 10-k resistor in the circuit using mesh analysis.

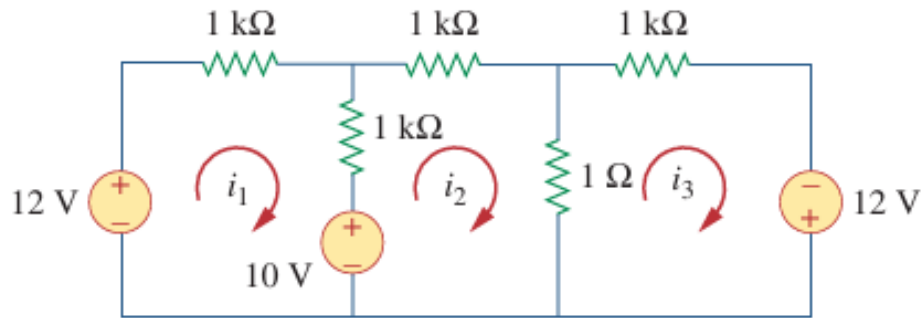


8. Determine voltages v_1 and v_2 in the circuit using mesh analysis.



Tutorial

9. Find the mesh currents i_1 , i_2 , and i_3 in the circuit.



10. Which of the circuits is planar? For the planar circuit, redraw the circuits with no crossing branches.

